Fredy Gunawan

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SUMMARY

A software engineer with over six years of experience, most of it building scalable and reliable backend systems in Go. Led the development and launch of new ads features at ByteDance and Tokopedia, with measurable gains in CTR, CVR and revenue. Experienced in leading cross-country collaboration, service migrations and on-call ownership. Led a backend team of three engineers at Kulina and mentored engineers as a part-time backend lead. Now looking to grow into tech lead and engineering manager roles.

WORK EXPERIENCE

Software Engineer Backend *February 2024 - May 2026*

ByteDance — *A global internet technology company operating a range of content platforms, the owner of TikTok.*

Responsibilities

- Directed development of a personalized ad delivery system based on aggregated user behavior.
- Integrated Tokopedia data with TikTok ad models, improving recommendation effectiveness.
- Designed and implemented a scalable data pipeline for TikTok Shop Ads platform.
- Led cross-functional, cross-country collaboration for platform integration.
- Owned on-call response, stabilization, and recovery of a mission-critical microservices system.
- Tech stacks: Go, C++, Python, Redis, Hive.

Achievements

- Consolidated CTR/CVR ranking models, saving ~2TB storage while sustaining CTR (+1.69%) and CVR (+0.18%).
- Developed ItemCF Recall and contributed to fine-ranking CTR strategy for Store Ads, achieving 5% ads GMV and 5% ads revenue contribution.
- Enabled GMV Max for Toko through experiment rules and VO model readiness, delivering 33.15% ads revenue contribution.
- Supported Invisible Demand experiments, contributing to 24.40% ads revenue.

Software Engineer Backend *January 2022 - February 2024*

Tokopedia — *One of Indonesia's leading marketplaces, serves 200 million users nationwide, contributing to 1.5% of the national economy.*

Responsibilities

- Implemented keyword validation mechanism to improve ad relevance and bidding effectiveness.
- Planned and executed ads service migration, infrastructure optimization, and rollout monitoring.
- Monitored, investigated, and resolved data pipeline issues impacting keyword insight quality.
- Collaborated cross-functionally to stabilize downstream ad performance metrics.
- Tech stacks: Go, Java, MySQL, ElasticSearch, Redis, GCP.

Achievements

- Improved CTR (+2.96%), CVR (+3.37%), auction coverage (+0.9%), and CPC (+13.7%) through keyword quality enhancements.
- Strengthened keyword insight reliability, increasing keywords in auction (+5.22%), impressions (+11.36%), clicks (+9.16%), and orders (+19.60%).
- Reduced annual infrastructure cost by ~30% and lowered long-term risk via service decoupling.

Software Engineer Backend, then Backend Lead *September 2020 - January 2022*

Kulina — *A food-tech startup offering an online food catering platform that delivers daily, customisable meal plans to consumers and businesses.*

Responsibilities

- Architected and developed the full backend system of food.id, including microservices, APIs, and databases.
- Led the backend team of three engineers for the last six months.
- Collaborated with frontend, QA, product, and operations teams for end-to-end delivery.
- Managed project scope, technical requirements, and delivery timelines.

- Tech stacks: Go, MySQL, MongoDB, Redis, Elasticsearch, NSQ, websocket, AWS (ECS, Elastic Beanstalk, S3), Travis CI.

Achievements

- Delivered a production-grade quick-commerce backend, enabling rapid launch and scalable operations.
- Improved system reliability and reduced infrastructure costs via service migration and optimizations.

Software Engineer Frontend *January 2020 - April 2020*

Software Usaha Global Indonesia — *A software house handling diverse client projects and building its own products.*

Responsibilities

- Developed client-focused applications and an in-house billing system.
- Tech stacks: PolymerJS, CSS.

Fullstack Web Developer *October 2019 - December 2019*

BitZero Informatika — *An IT consulting company that builds software solutions required by its clients.*

Responsibilities

- Acted as IT consultant, gathering client requirements and aligning technical feasibility with business goals.
- Developed custom applications based on the agreed-upon specifications from the consultation.
- Tech stacks: PHP, JS, CSS, MySQL.

SIDE PROJECTS

Backend Lead (Part-Time) *December 2023 - May 2024*

Wuiiy — *A skincare platform providing personalized consultations, prescriptions from doctors, and in-app purchase of recommended products.*

Responsibilities

- Led backend development for a cutting-edge skincare platform offering personalized consultations, doctor prescriptions, and seamless in-app product purchases.
- Led Scrum processes using Agile methodology, prioritizing high-impact tasks to accelerate feature delivery.
- Filtered and refined backend product requirements, preventing team overload while ensuring key features were implemented efficiently.
- Mentored junior engineers and optimized workflows, enhancing system performance for a growing user base.

EDUCATION

Universitas Gunadarma 2015-2019

Bachelor's in Computer Science

SKILLS

- **Backend & Microservices:** Design, development, and maintenance of mission-critical systems, including ad delivery pipelines and data processing platforms.
- **Advertising Systems & Optimization:** Keyword insights, CTR/CVR modeling, ad bidding optimization, and campaign performance improvement.
- **Data Analysis & Experimentation:** Analyzing experimental results and advertising metrics (CTR, CVR, CPC, GMV) to inform release decisions.
- **Data Engineering & Pipelines:** Monitoring, debugging, restoring pipelines; cross-platform data integration; service migrations.
- **Project & Program Management:** Planning and executing migrations, tech improvements, and platform rollouts; coordinating cross-functional teams.
- **Team Leadership:** Leading a backend team of three engineers, running Scrum, and mentoring engineers.
- **Collaboration & Stakeholder Management:** Working with backend, QA, algorithm, and external teams; translating client requirements into effective technical solutions.